Four Choices Behind One Number: Reporting the Incremental Value of a Recorded Field

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Abstract

Papers, business cases and tool evaluations report what a recorded field is worth as a single number. That number is a function of four choices which are almost never stated: the *baseline* of already-recorded fields the comparison is allowed to contain, the *metric*, the *operating point*, and the *population* of the register the field reads from. We write the quantity as $V(f \mid B, m, \theta)$ and propose that a feature-value claim be reported as a surface over those arguments rather than as a point.

The demonstration is a configuration management database (CMDB) on a public event log of 45,455 incidents from a bank's IT operation, predicting at intake whether an incident will be reassigned. Each choice moves the answer by more than the effect the previous version of this paper reported. Admitting one further field the organisation already records — which group logged the incident — takes the item's value from +0.183 AUC to +0.103. Admitting a second, which a file we had not obtained now shows is written by the service desk before the incident exists, takes it to +0.001 $[-0.002, +0.003]$: the headline does not survive our own admissibility criterion once the evidence to apply it is in hand. Under six defensible instruments on the same scores the reduction runs 43.7% to 60.3%, and the AUC figure we published is the smallest of them; under net benefit it runs 6.3% to 119.2% across the operating range, and the item is resolvably harmful in a band above the base rate. At half population the item is worth +0.082 if the register's long tail is missing and +0.064 if its core is.

We pre-registered a generic protocol and applied it to 22 public event logs across seven domains. It admits 13; the reduction is resolvably positive on

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three. The generality claim we registered is therefore falsified, and we report the negative, the exclusions and the condition that does govern: on most logs the entity is worth nothing to the intake baseline, so there is no value for a free field to absorb.

Eleven errors of our own are reported as results rather than edited away, three of them found in the round that produced this version. Two are claims whose every individual number was correct and whose asserted relation between them was not.

Keywords: incremental value, baseline specification, evaluation metrics, predictive process monitoring, configuration management database, variable importance, decision curve analysis, pre-registration

1. Introduction

A business case for a data programme is a comparison, and the second arm of that comparison — performance *without* the data — is a choice the analyst makes, usually silently. So is the metric the comparison is stated in, so is the operating point at which that metric is read, and so is the assumed completeness of the register being bought. Each of those choices is routinely left implicit, and the result is reported as one number.

We write the quantity such a claim actually names as

$$V(f \mid B, m, \theta), \tag{1}$$

the incremental value of a feature f over a baseline feature set B, under metric m, at operating point θ; and for a feature that is a lookup into a register we treat that register's population as a fourth choice, one that acts on f rather than on the comparison. The claim of this paper is not that V depends on those arguments — that is a theorem, and Section 3 says whose — but that on a real decision, with every choice held to a defensible range, the dependence is larger than the effect being reported. We demonstrate it by taking our own previously published headline apart along each argument in turn. One of those demonstrations removes the headline.

The demonstration runs on a configuration management database (CMDB): the register of items an IT estate contains, bought to make operational analytics possible, evaluated here on a named task — predicting at intake whether an incident will be reassigned — on a public log of 45,455 incidents from a bank's IT operation.

This paper reports four things.

1. **Every choice moves the answer, and the baseline choice removes it.** Admitting one further field the organisation already records — which group logged the incident — takes the item's value from $+0.183$ to $+0.103$ $[+0.094, +0.113]$ AUC. Admitting a second, the knowledge-article reference, takes it to $+0.001$ $[-0.002, +0.003]$. The previous version declined to admit that second field because it could not establish when the value was written; a file from the same public collection, which we had not obtained, settles where it comes from (Sections 6 and 13).
2. **The metric and the operating point are choices of the same kind.** On identical scores, six defensible instruments put the reduction between 43.7% and 60.3%, and ROC AUC — the number the previous version printed — is the *smallest* of them. Net benefit, read at a single operating point as decision-analytic practice requires, puts it between 6.3% and 119.2% depending on which point, and the item is resolvably harmful in a band above the base rate. We identify which mechanism produces which disagreement (Sections 8 and 9).
3. **A pre-registered protocol over 22 public logs, and a falsified generality claim.** We registered a generic procedure, a corpus and an exclusion rule before running any of them. The rules admit 13 logs across six domains; the reduction is resolvably positive on three. The registered claim that the effect is a property of process event logs rather than of ITSM data is therefore *falsified*, and Section 11 reports the negative together with the condition that does govern: on most logs the entity is worth nothing to the intake baseline, so there is no value for a free field to absorb.
4. **Eleven corrections, reported as results.** Eight were found in earlier rounds; three in the round that produced this version. Two of the three are claims in which every individual number was correct and the relation the prose asserted between them was not (Section 15).

What kind of contribution this is.. We report no deployed system. This is a retrospective study on two public benchmarks, offered as an evaluation practice. Section 9 states what the choice costs in units closer to those a service desk operates in, and Section 5 states which *layer* of configuration data carries the value, which is the question a buyer actually faces.

On what this paper used to claim.. Earlier versions reported larger findings withdrawn under adversarial review, each an artifact of a control we built ourselves. Section 15 lists eight corrections that survive here as reported results rather than silent edits, because each is the kind of error this paper is about. Two were found in the round that produced this version, and one of them — a claim about a public dataset that our own repository contradicted — had stood through eight rounds of review.

2. Background

Prediction on incident event logs.. Our task sits at the boundary of predictive process monitoring, surveyed and benchmarked across real-world logs [24, 25] on data of the kind process mining formalises [1]. We predict at $t = 0$ from creation-time attributes with no prefix, which is tabular classification on a process log rather than prefix-based monitoring in the sense those benchmarks evaluate; we note that Teinemaa et al. [24] explicitly excluded the 2013 and 2014 BPI Challenge logs from their benchmark for lacking a static/dynamic attribute split. The BPI Challenge logs are the standard public instances [8, 22]; comparable exports exist elsewhere [3]. Such logs carry documented quality hazards [23]. One hazard bears directly on us and we state it as our own measurement rather than attributing it: values recorded once per record can be replicated onto every event of that record, which makes constancy across a case uninformative about when the value became known. Establishing what a resource stamp on an event denotes is the business of organisational mining [2], which is what Section 4 does from first principles for the field our result turns on. The state of the queue a case arrives into — the inter-case perspective — carries real predictive content in this setting [21], and Section 6 admits four free creation-time congestion features to the baseline for that reason.

The CMDB and what justifies it.. A CMDB records configuration items and their relationships, and is the asset on which the configuration-management practice of ITIL and its successors rests. Survey work on ITSM framework adoption [15] finds that realised benefits grow with implementation maturity, but measures them as practitioner-reported outcomes rather than as the analytic value of the configuration data itself.

What is and is not already measured.. Prior peer-reviewed work uses the affected configuration item as one input among many without isolating its value. Schad et al. [20] predict ticket reassignment — our own target — with a relational graph convolutional network on a public ServiceNow log, with the configuration item among the incident-node features. Sarnovský and Surma [19], working on the same Rabobank log we use, report ROC AUC and rank CI type, CI subtype and service component among the most significant attributes. Kapel et al. [11] predict change-induced incidents at a large bank and rank CI Name third by SHAP importance, behind two free-text fields. None of the three performs a controlled with-and-without comparison attributing a change in a fixed metric to configuration-item identity against a stated baseline, which is what we do; but the reader should note that the question is not untouched, and that in the one study which ranks a free-text field against the configuration item, free text wins.

The methods a CMDB is bought to enable.. Surveys of AIOps for failure management [30, 17, 16] catalogue triage, failure-prediction and root-cause methods that take configuration data as input. Incident triage in particular — deciding which team owns an incident — has been treated as a learning problem on operational logs at industrial scale [5], as has failure prediction across a cloud estate [14].

What the comparison is measured against.. The value of a feature *given* a baseline is what algorithm-agnostic variable-importance frameworks formalise; Williamson et al. [28] develop this for R^2-type predictiveness and Williamson et al. [29] extend it to AUC, which is our metric. Our overlap is the redundancy that additive importance measures share credit for [7]. Section 3 says what this paper adds to those frameworks and what it does not.

The estimand itself — the change in a fixed metric when one variable joins an existing model — has been argued over in biostatistics for two decades, and that literature reached our conclusion first. Cook [6] shows that an increment in AUC is a poor measure of a variable's incremental value and depends heavily on the model it is added to. Vickers and Elkin [26] supply the principled replacement for the capacity-limited framing we used in earlier versions and withdraw in Section 9. We claim no priority over these; we report that the failure mode they describe is, on operational data, large enough to change a procurement decision.

Within machine learning, Rendle et al. [18] and Ferrari Dacrema et al. [9] show insufficiently tuned baselines accounting for reported gains in recommender systems; benchmark conclusions are sensitive to variation that is seldom reported [4], and documentation of the choices producing a result is thin [10]. Where the choice concerns which fields may enter, it shades into leakage [13, 12], whose formal treatments accommodate admission rules as constraints set by the problem definition rather than derived from it. In predictive process monitoring, Weytjens and De Weerdt [27] show that avoiding a different leak — case prefixes shared between training and test under naive temporal splitting — needs explicit construction; we adopt their strict temporal splitting and, at their prompting, examine the extract boundary in Section 4. We report a field-admission failure mode for a deployment decision rather than a leaderboard, where it is not a modelling detail but the number the business case turns on.

3. This Is Not Just Conditional Variable Importance

3.1. The estimand

Equation 1 names $V(f \mid B, m, \theta) = m(B \cup \{f\}, \theta) - m(B, \theta)$. For two nested baselines $B_0 \subset B_1$ we report the *admissibility reduction*

$$R(f \mid B_0, B_1, m, \theta) \;=\; 1 - \frac{V(f \mid B_1, m, \theta)}{V(f \mid B_0, m, \theta)}, \tag{2}$$

the share of a feature's apparent value already carried by fields the organisation records for nothing. θ is absent for metrics that integrate over the operating range; Section 8 says which those are. We report R only where $V(f \mid B_0, m, \theta)$ is positive in every bootstrap draw. A ratio whose denominator crosses zero is not a quantity, and this paper has previously printed one that did.

Four independent choices therefore sit inside any single-number claim: which already-recorded fields B contains, which m, which θ, and — for a feature that is a lookup into a register — how completely that register is populated. Sections 6, 8, 9 and 10 vary one at a time on one dataset, holding the cohort, the split, the estimator and the seed fixed throughout, so that any movement is attributable to the choice and to nothing else. Section 12 states what we propose be reported in place of a point.

3.2. What is and is not new here

The objection we expect first is that measured importance is baseline-relative, that Williamson et al. [29] and Covert et al. [7] formalise exactly that, and that a study reporting it on one log is reporting a known fact with a new number attached. The objection is correct about the formal content and we concede it in those terms. Four things are added here that no formal framework supplies.

A magnitude on a named decision.. Those frameworks establish that a feature's importance is defined only relative to a baseline. They do not say how large the dependence is for any particular feature on any particular decision, and a procurement decision turns on the size. On this task the answer is that the choice of baseline moves the measured value of an entire data programme by roughly half, and that half is the difference between an investment case and none.

The right estimand is not an average over baselines.. A Shapley-style attribution averages a feature's contribution over subsets of the others, which is the correct thing to compute when the question is how to apportion credit within a fixed model. It is the wrong thing to compute here. A business case needs a feature's value against *the baseline the organisation will actually have* — the fields already in its ticketing tool — not an average over baselines it will never be in. We therefore report a ladder over named, admissible baselines rather than a single attribution, and Section 4 states the admissibility criterion before applying it.

The overlapping field is free.. The interesting case is not that some baseline choice changes the answer; any correlated feature does that. It is that the field which absorbs half the measured value is one the organisation *already records at no cost*, sitting in the same export, requiring no programme and no budget line. That is what makes the analyst's choice invisible rather than merely arbitrary: nothing in the data flags the omission, and the omitted field has no price tag to prompt a question about it.

The metric argument is not in those frameworks at all.. Williamson et al. [29] and Covert et al. [7] are stated for a fixed predictiveness measure. Nothing in them says that two defensible measures applied to the *same scores* disagree about how much of a feature's value a second feature absorbs, still less by how much. That is an empirical question, Section 8 answers it here, and the answer is that on this decision the metric choice moves the reduction by more than the free field does.

Where it appears and where it does not, measured rather than assumed.. Baseline-relativity is a theorem; whether a particular pair of fields overlaps in a particular estate is an empirical question with no general answer. We pre-registered a protocol and applied it to 22 public logs (Section 11) rather than generalising from one. The answer is mostly negative and is reported as such.

What we do not claim.. We do not claim a new estimator, a new importance measure, or priority over Cook [6], who stated the underlying point for AUC in 2007. Nor do we claim any of the four choices is illegitimate: each is correct about the question it asks. The contribution is that those are different questions, that a single number answers one of them without saying which, and that on this decision the spread between them is larger than the effect.

4. Data and Task

4.1. The primary organisation

We use the incident records of the BPI Challenge 2014 log [8] from Rabobank Group ICT, recorded in HP Service Manager, with the activity log shipped alongside. The configuration-item column we use is `CI Name (aff)`, the item affected at the time the incident was recorded, not the closure-time `CI Name (CBy)`.

Cleaning.. Of 46,809 rows, 203 carry no incident identifier and one an unparseable timestamp. A further 1,150 opened before 2013-10-01 are left-censored — already open when the extract began — and are reassigned at 81.2% against 40.0% for the incidents we keep. Removing them leaves **45,455** incidents to 2014-03-31; a temporal 70/30 split gives 31,818 training and 13,637 test, positive rates 0.413 and 0.372. The cutoff is not taken from the outcome: monthly volume rises from 857 incidents in September 2013 to 8,606 in October. The extract is right-censored too — incidents opened near its end cannot have long lives — so we re-measured on cohorts truncated up to a month early; the shrinkage moves between 42% and 45%, so the boundary does not drive it. The affected-item field is 100% populated across 2,929 items, 2,554 seen in training. That population rate is a property of the extract and of a tool configuration that copies the item onto the incident from the originating interaction record; it is not evidence about CMDB accuracy in general, and Section 14 returns to the point.

Task and estimator.. Predict at creation whether an incident will later be reassigned. We call the target reassignment rather than misrouting throughout: it fires on 37.2% of test incidents, which is routine handling and not error, and Section 9 reports the ladder at stricter thresholds. Median handling time rises from 1.59 hours for incidents never reassigned to 47.53 hours for those reassigned five or more times; we use this to motivate the task, not as a recoverable saving.

The intake block is four fields carried on the incident form: Category, Impact, Urgency and Priority. It is effectively three. `Priority` is a deterministic function of `(Impact, Urgency)` on this log — 19 occupied cells, none carrying more than one Priority, across all 46,809 rows — so it adds no information, and dropping it moves intake AUC from 0.562 to 0.564. Sarnovský and Surma [19] reported this redundancy on this same log in 2018. We keep the column because it is what an analyst building the naive arm would include, and we disclose it because a paper arguing that baselines are under-specified should not ship an under-specified baseline in silence. The four fields together take only 23 distinct combinations in training and 23 in test, which is a fact Section 9 needs and which no choice of estimator can change.

The primary estimator is one-hot logistic regression, chosen because it is bin-free: scikit-learn's `LogisticRegression` with L2 penalty, inverse regularisation strength $C = 1.0$, the `lbfgs` solver and a cap of 3000 iterations, with `OneHotEncoder` set to ignore unseen levels. The penalty is held fixed across every arm; tuning it per arm on an inner split of training data moves the headline reduction from 43.7% to 43.2%. Two further estimator specifications are reported in Section 6. Encoders and rankings are fit on training data only; intervals are 2,000-resample paired bootstraps, except Table 4, whose counts are re-ranked in each draw and use 400. Monte-Carlo nulls use 15 to 100 draws, the count set by cost and recorded in the code that produces each. Every sampling script is seeded from a single constant, so the reported figures are reproducible exactly.

The free field, and what it is not.. The activity log records, on the `Open` event of every incident, an assignment group: 49 in training, none missing. An earlier draft of this paper called it the queue the service desk routed the incident to. That was an assertion, not a finding, and it is wrong. Three checks say so. The `Open` rows carry 50 distinct groups where the `Assignment` rows carry 218, and a routing destination cannot be less diverse than the teams that then do the work. One group accounts for 67.0% of `Open` rows but only 18.4% of all activity rows. And the `Open` group matches the group on the incident's first `Assignment` activity for just 15.1% of the 37,577 incidents that have one. The field is the group that *logged* the incident: one central desk for two thirds of tickets, a long tail for the rest. We do not characterise that tail — its openers match the first `Assignment` group only 21.8% of the time and appear anywhere in the later work rows only 59.1%, so calling them teams opening their own work would be another assertion of the kind we are correcting. It is also concentrated and unstable: the largest group holds 62.1% of training incidents, and the number of live groups falls from 49 to 32 between our two halves, while in the test half it logs 78.6% of incidents.

The free field is not a restatement of the target.. A reader may reasonably suspect the opposite of what Section 7 argues: that a field naming a *group* cannot help but encode a target defined as a *change of group*, making the whole ladder circular. That reading makes a directional prediction, and the data contradicts it. The central desk logs most tickets and does a small minority of the work, so on the circular reading it should be the one handing work on, and should reassign more than everyone else. It reassigns *less*: 0.309 against 0.603, a difference of -0.294 $[-0.315, -0.275]$. What the field separates is a low-reassignment intake channel from a high-reassignment one. Nor is it a strong predictor in its own right: the per-group outcome rate, applied as a lookup with no model, scores 0.642, and the one-bit central-desk contrast 0.606. A field that restated the target would score near the ceiling.

Why not the real routing decision.. The obvious repair is not available. The first `Assignment` group occurs strictly after `Open` on every incident that has one, a median of 46 minutes later, and 7,878 of 45,455 never have one. It fails the criterion below, and substituting it would introduce the leakage that criterion prevents.

Admissibility, and mutation.. A field on the `Open` row is admitted only if it is a per-event observation, tested by whether it varies across an incident's activity rows; the opening group varies for 92.56% of cohort incidents and matches the last-observed group for only 21.35%. Because the incident file is a closed-record snapshot we also report how often each field is edited after creation: Urgency on 1,107 incidents (2.44%), Impact on 1,084 (2.38%), the affected item on 159 (0.35%). Incidents whose Impact was edited are reassigned at 0.66 and those whose Urgency was edited at 0.67, against 0.39 for the rest, so this is not neutral; Section 6 reports the headline with those fields dropped and with the edited incidents removed. The activity log carries no Category or Priority change type, so for those two fields the snapshot cannot be audited at all; our disclosure covers the fields for which an audit exists, not every field we use.

A second criterion governs the other direction, and Section 5 needs it: a field may join the CMDB-free baseline only if the organisation would hold it *without* the CMDB. That excludes every field which is a function of item identity, because such a field exists only where the item register does.

4.2. The second organisation

Until this round we described this study as single-organisation of necessity, on the ground that of the three public ITSM incident logs in common use, one records the affected item on every incident, the second on 0.2%, and the third had no such field at all. The third clause was false, and our own repository contained the evidence: the BPI Challenge 2013 incident log [22] from Volvo IT, recorded in VINST, carries a `product` attribute with 704 distinct values, exactly one per trace, on all 7,554 traces and none missing. It is a coarser identifier than an instance-level CI name, but it is an affected-item field, and the challenge's own headline question was ping-pong behaviour between teams — the same outcome our primary target measures.

We therefore run the ladder there too. The item analogue is `product`; the free field is the `org:group` on the first event, of which there are 338; the intake block is `impact`, the one intake-form field the two logs share. The target is at least one change of `org:group` along the case, which fires on 50.6% of cases, with a stricter variant at two or more changes (27.2%). We use the same temporal 70/30 split and the same estimator.

One difference cuts against us and we state it rather than let a reader find it. On BPI Challenge 2014 the opening group matches the incident's first `Assignment` group only 15.1% of the time: it is the desk that logged the ticket, a different object from the routing sequence that defines the target. On BPI Challenge 2013 the opening group *is* the first term of the `org:group` sequence whose variation defines ping-pong. The coupling is therefore strictly tighter there, the free field is correspondingly worth more (+0.188 against +0.082), and the Volvo reduction should be read as an upper bound rather than as a second draw from the same distribution.

Grouping	levels	AUC	gain
CI Type (aff)	13	0.671	+0.027
CI Subtype (aff)	61	0.657	+0.013
Service Component WBS (aff)	256	0.722	+0.078
CI Name (aff)	2,554	0.748	+0.103
CI Name, marginal over the service component	—	—	+0.023

Table 1: At what resolution configuration data carries its value, over a baseline that already contains the opening group. Levels are counted in the training half.

5. Which Layer of Configuration Data Pays?

The field-admission question of Section 6 has a companion that is more directly a buying decision, and it is where this paper's novel finding sits: not whether configuration data is worth anything, but at what resolution, and whether what pays is the configuration *attributes* at all. Four nested groupings of the same estate are all 100% populated and free once the item is known. Measured over the intake-plus-group baseline of 0.644:

A 256-way service-component grouping captures three quarters of what instance-level identity is worth, and instance identity adds +0.023 over it. The classification layers proper — CI type and subtype — carry little, and CI Subtype scores below CI Type despite being finer, which is what a fixed penalty does to a sparser column. This is the same +0.023 that Section 7 reports for admitting the service component to the baseline, seen from the other side: the two are the same quantity, and a reader who regards an application or service register as something an organisation has without a configuration-management practice should read the CMDB's marginal contribution here as that figure.

Why these fields are in the ladder and not in the baseline.. The obvious question is why CI Type and CI Subtype, being 100% populated and free, are not simply admitted to the CMDB-free baseline alongside the opening group. The answer is the second admissibility criterion of Section 4 and it needs no model: both are exact functions of item identity. Neither varies on a single one of the 2,929 items, so both exist only where the item register exists, and admitting them would be admitting the CMDB to the CMDB-free baseline. The service component is the case this argument does not dispose of, and the quantitative gap is instructive: it varies on 58 of 2,929 items, carrying 8.7%

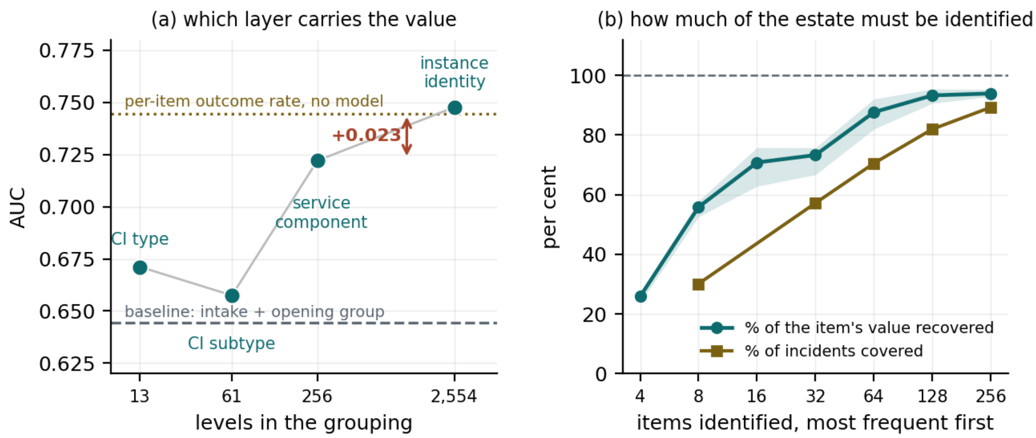

Figure 1: (a) Which layer carries the value, over a baseline that already contains the opening group. The dotted line is a per-item outcome rate applied as a lookup, with no model and no configuration attribute. (b) How much of the estate must be identified: the share of the item's measured value recovered by identifying only the k most frequent items, against the share of incidents those items account for. Band: min–max across five split points.

of incidents, where the opening group varies on 565 items carrying 92.5%. The line between the two fields is real and it is a matter of degree, which is why we report the headline both ways rather than choosing.

The signal is an outcome history keyed on identity.. A per-item reassignment rate computed on the training half and applied to the test half as a lookup table — no model, no other field, no configuration attribute of any kind — scores 0.744, against 0.745 for item identity alone under the full estimator and 0.748 for the paper's complete model. Whatever the CMDB supplies here, it is not the item's attributes; it is a stable identifier under which six months of outcomes can be accumulated. The defensible reading is that maintaining such an identifier across a changing estate — dedupe, reconciliation, retirement — is itself a configuration-management function, and we think that reading is right. But it is a different claim from "the CMDB is worth +0.103", and an organisation that can key its ticket history on any stable token can have most of this without one.

Estate concentration.. Required scope is bounded by a property of the estate needing no model: the top 8 items generate 30.2% of training incidents, the top 64 70.5%, and the top 128 — 5.0% of the training vocabulary — 82.0%,

Organisation	Baseline	AUC	+ item id.	gain [95% CI]
Rabobank	intake only	0.562	0.746	+0.183 [+0.172, +0.195]
(BPIC 2014)	+ opening group	0.644	0.748	+0.103 [+0.094, +0.113]
Volvo IT	intake only	—	—	+0.238
(BPIC 2013)	+ opening group	—	—	+0.092

Table 2: The value of knowing the affected configuration item, with and without a field the organisation already records, on two organisations. The Volvo rungs use a coarser item identifier and a more tightly coupled free field; see Section 4.2.

all from a frequency count before funding anything. Identifying only those items recovers most of the value. Averaged over five split points, the top 8 recover 56% of the +0.103, the top 64 recover 88% and the top 128 recover 93%; across those splits the three range over 52–58%, 81–93% and 90–96%. These are min–max spreads over a design choice, not bootstrap intervals, and we write them as ranges to keep the distinction. We give the mean and range rather than one split's curve because a single curve is not monotone at this resolution: at $k = 32$ the across-split spread is 9 points. The same holds with the group removed from the baseline — 89% at $k = 64$ — so this is a property of the estate's concentration, not of our field-admission decision. Taken with the resolution ladder, the practical reading is that the marginal AUC of the 2,801 items outside the top 128 is small, and that is where the cost of a CMDB programme lives.

6. The Admissibility Effect

Against four intake fields the affected item looks decisive. Adding the opening group cuts its measured value by 44%. The gains are roughly 28 and 17 standard deviations outside a null in which the incident-to-item association is shuffled, preserving column count and marginal distribution while destroying the signal — we round because re-drawing the same null at a different draw count moves the first figure by about 0.7 — a comparison that matters because adding item identity adds 2,554 indicator columns, which is not free under a fixed penalty. These z figures pool the null's Monte-Carlo dispersion with the gain's own bootstrap standard error; dividing by the null alone, as an earlier draft did, would report 51 and 33. Because these nulls use as few as 15 draws, a two-digit z is an extrapolation from a dispersion

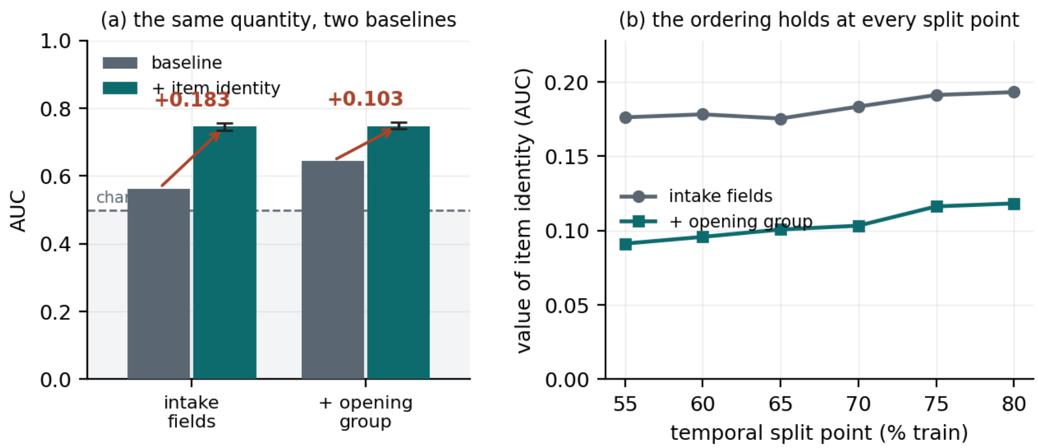

Figure 2: (a) The same quantity under two baselines, on the metric's full range; the whisker on each treatment bar is the paired bootstrap interval of that bar's own gain. (b) The ordering holds at every temporal split point from 55% to 80%.

estimate and not a tail probability; the load-bearing statement is that no null draw comes near either gain.

It is not a split artifact: across six split points the two gains range +0.175 to +0.194 and +0.091 to +0.119 (Figure 2b), and across split point, target threshold and cleaning cutoff the second ranges +0.067 to +0.130 — wider than any bootstrap interval, which conditions on all of those. Nor does the disclosed field editing explain it: reducing the intake block to Category alone gives +0.106, and restricting to the 44,227 never-edited incidents gives +0.107.

The reduction, over the whole design space.. The reduction rather than either gain is what we argue transfers, so it needs the same disclosure. Across split points it runs 38.7% to 48.3%; across the three target thresholds 43.6% to 48.3%; across the three estimator specifications 42.7% to 47.1%; and across five cleaning cutoffs from no censoring removal at all to a two-month truncation, 36.1% to 44.0%. Over the whole design space it runs **36.1**% to **48.3**%. The bootstrap interval of $[40, 48]$ we report at the published setting conditions on every one of those knobs, and an earlier version of this paper quoted it without saying so while asserting in its title that the reduction was nearly a half. At the low end of the design space it is closer to a third. What survives across the space is that the reduction is large and its sign never changes.

It is not congestion.. A reader from the process-mining community will ask whether the affected item is standing in for load, since the inter-case perspective is known to carry real content on this kind of log [21] and neither of our baselines looks at the state of the queue an incident arrives into. Four features are free at creation and are computed from the open and close timestamps of *other* incidents, every one of which is in the past at the moment of creation: the open backlog, arrivals in the preceding hour, hour of day and day of week. Continuous features are cut at training deciles, so the estimator is unchanged. Admitted to the baseline on the same criterion as the opening group, they move the item's measured value from +0.103 to +0.100, and the reduction from 43.7% to 45.7% $[41, 50]$. Alone, without the intake block or the group, the same four features score 0.497 — at this horizon, on this log, creation-time congestion carries essentially nothing. The item is not a load proxy.

It replicates on a second organisation.. On Volvo IT the same ladder gives +0.238 against intake and +0.092 once the opening group is admitted, a reduction of 61.3% $[54, 68]$; at the stricter two-change threshold, +0.249 and +0.139, a reduction of 43.9% $[31, 55]$. Neither interval includes zero. Across six temporal split points the first runs 60.3% to 67.8% and the second 40.8% to 53.8%. The qualitative claim — that admitting one field the organisation already records removes a large fraction of the item's measured value — holds on a second organisation, a second tool and a second country. The magnitude does not transfer, and Section 4.2 gives a structural reason to expect the Volvo figure to be the larger of the two.

It is not an artifact of the estimator.. The dimensionality null rules out the regularisation burden of 2,554 sparse columns, but a reader may want the effect shown where that burden cannot arise. We re-represent item identity as a single cross-fitted target-encoded column — fitted on training only, out-of-fold, so no row sees its own outcome — which makes the item one continuous feature and makes histogram boosting usable on a variable that otherwise collapses 2,554 items into 137 bins. Across one-hot logistic regression, logistic regression on the encoded item, and boosting on it, the first rung ranges +0.173 to +0.184, the second +0.091 to +0.104, and the reduction 42% to 48%. Encoding a *shuffled* item column under the same pipeline returns -0.0001 ± 0.0016 and $+0.0042 \pm 0.0020$ on the group-aware rung. The first collapses to zero; the second does not, and it is eleven standard errors from

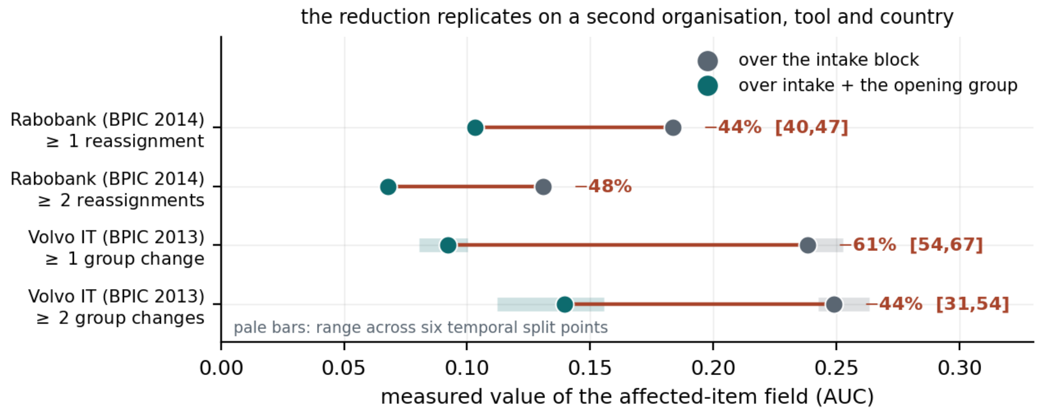

Figure 3: The ladder on both organisations, at two target thresholds each. Each segment runs from the item's value over the intake block to its value once the opening group is admitted; the label is the reduction, with its paired bootstrap interval where one was computed. Pale bars are the min–max range across six temporal split points, which was computed for the Volvo rungs only.

zero, so it is a real systematic effect rather than noise and it must be bounded on both rungs and not one. Run on the intake rung as well, the same control returns $+0.0002 \pm 0.0015$ and -0.0036 ± 0.0025. Subtracting each rung's own residual from that rung's gain moves the reduction from 42.7% to 42.6% under the encoded logistic model and from 47.1% to 50.5% under boosting. The correction is therefore worth at most 3.4 percentage points and, where it bites, it makes the reduction larger.

Three target definitions.. Because reassignment is common, we repeat the ladder at stricter thresholds. Requiring two or more reassignments (21.8% of incidents) gives $+0.131$ and $+0.068$; three or more (10.6%) gives $+0.151$ and $+0.080$. The magnitude is threshold-dependent and not monotone, but the interval on each +group rung excludes zero (we compute intervals for that rung only), the ordering never reverses, and the reduction stays between 43% and 49%.

Does it survive a change of task?. A gap on one target may be a property of it, so we repeated the ladder on two further outcomes within the primary organisation. *Reopening* is close to an independent failure mode: 2,096 incidents, correlating with reassignment at $+0.14$. *Long handling* — handle time above the training third quartile — correlates at $+0.40$, so it is corroboration only.

On reopening, item identity is worth +0.083 against the intake block and +0.055 once the group is admitted, a reduction of 33%; on long handling, +0.118 and +0.078, a reduction of 34%. Every rung sits outside its matched-dimension null. But the reduction is what we argue transfers, so it needs its own interval, and paired bootstraps give $[40, 48]$ on reassignment, $[28, 38]$ on long handling and $[-1, 60]$ on reopening. On the one target we argue is near-independent the reduction is *not* resolvably different from zero ($P \leq 0$ is 0.03), and long handling, correlated at +0.40 with the primary target, cannot carry a replication alone. Reopening is rare and its gains stand only 5.5 and 4.2 pooled standard deviations from their nulls.

Cross-target replication within one organisation therefore fails to resolve, while cross-organisation replication succeeds. We take the second as the stronger evidence, because the objection it answers — that the effect is a property of one estate — is the one a reader should press hardest, and because both Volvo intervals exclude zero. The reduction is largest on the target the opening group is most directly about, which is consistent with Section 7 rather than predicted by it: the mechanism says nothing about how the reduction should order across targets.

7. Where the Difference Goes

The two rows of Table 2 differ by 0.080, which is not itself informative: for any two variables, the drop in one's gain when the other joins the baseline equals the drop in the other's. Read four ways, the item is worth +0.183 over intake and +0.103 once the group is present; the group is worth +0.082 over intake and +0.002 once the item is present.

This section is weaker than the version we published previously, and the weakening is a finding rather than a concession. Of the four legs that supported the overlap reading, one has been withdrawn as an algebraic identity, one no longer resolves once its null is matched properly, and two survive.

What survives, first.. Adding the opening group to a model that already knows the item is worth +0.0017 $[+0.0001, +0.0034]$. It clears a matched-dimension null of -0.0009 ± 0.0005 that no draw reaches, so it is not the cost of adding columns; but across split points and penalties it ranges +0.0002 to +0.0072, so the honest statement is that its unique contribution is *under* 0.01 *AUC and not resolvable more finely.* Because the real column is nearly constant within an item while the null's is drawn per row, this null is matched on dimension but not on that collinearity, and a sceptical reader should treat the interval as the weaker of the two claims.

What survives, second.. Item identity does not determine the group, but predicts it far better than chance: a modal-group lookup on item identity reproduces which group logged the incident on 90.3% of test incidents against 78.6% for always guessing the largest, while class-balanced accuracy reaches only 34.1%. Item identity resolves the coarse contrast — central desk or specialist team — and much less of the fine one. The two variables share 1.47 bits of mutual information.

A margin that was a knob.. Randomising the group label *within* each item — which destroys the label's own identity while preserving what the item implies about it — retains 91% of the group's +0.082 gain. We previously compared that against a floor built by randomising within a random partition *of items* at 49 cells, matching the opening group's own cardinality, which retains 42%, and reported the 49-point margin as evidence that item identity specifically was doing the work.

That comparison is not matched. The real leg partitions on item identity, which is 2,929 cells; the floor was drawn at 49. Retention rises monotonically with the fineness of the partition, so the margin is a function of a knob that was set to the value maximising it. The stated ground for stopping the sweep — that at one cell per item the null becomes the real thing — is wrong, because a *random* partition of n items into n cells is not the identity partition and stays routing-blind. Swept out to matched granularity the floor retains $87.5\% \pm 3.6\%$ against the real leg's $91.0\% \pm 1.6\%$: a margin of 3.5 points at $z = 0.9$, against this project's own resolvability bar of $|z| > 3$. We therefore withdraw the margin. What the sweep shows is that retention is high at fine granularity whether or not the cells mean anything, which is a fact about granularity and not about routing.

A withdrawn measurement: the asymmetry was an identity.. We previously reported that, without any estimator, item identity carries 60.4% of the opening group's information while the group carries 19.6% of the item's, and read the asymmetry as evidence that the mechanism runs from item to group. The reading is unsupported and the measurement is circular. The uncertainty coefficient $U(A \mid B) = I(A; B)/H(A)$, and mutual information is symmetric, so the ratio of the two coefficients is identically $H(\text{item})/H(\text{group})$ — a function of two marginal entropies and of nothing else. On this log that ratio

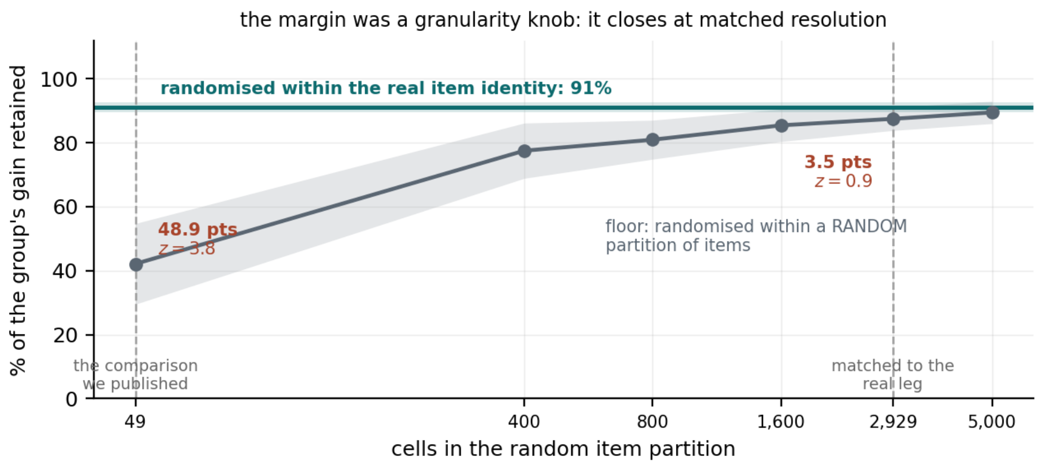

Figure 4: Randomising the group label within each item, against the same randomisation within a random partition of items, as a function of how fine the partition is. The margin we published was measured at 49 cells; the leg it bounds partitions on 2,929. It closes as the partition approaches the granularity of the real leg.

is 3.09, and the two coefficients, their shuffled floors, and the floor-corrected excesses we published all stand in that same ratio to six decimal places. Any two variables whose entropies stand in that ratio produce those two numbers, whether or not either determines the other. We withdraw the asymmetry and the directional claim it supported. The mutual information itself, 1.47 bits, is a real measurement and is reported above; its *direction* is not measurable this way.

The reduction is graded in the field's resolution.. Coarsening the group and re-running the ladder gives reductions of 27%, 28%, 36% and 44% at two, four, eleven and 49 levels. This is a consistency check rather than an independent prediction, because at each resolution the reduction is close to that coarsened field's own gain over intake — by the identity above, the same statement as the first measurement — and the two coarsest points have overlapping intervals, so only the trend is claimed. A single binary flag — central desk or not — already recovers 61% of the group's baseline gain and 63% of the reduction. The field that must be admitted is close to one bit, far cheaper to reproduce elsewhere than a 49-way taxonomy.

Is the field really free?. One objection goes to the heart of the reading. The opening group may be free only because a human at the desk already knew what the ticket was about — in which case it is undocumented configuration knowledge, and a baseline containing it is not a CMDB-free baseline. A third reading is at least as likely and we had missed it: the contrast may be an intake-channel indicator, separating an out-of-hours or overflow or automated route from the central desk, in which case it is neither independent of the item nor configuration knowledge held by a person. The one-bit result makes the question concrete without settling it: the contrast doing most of the work is the central desk against everything else, and Section 4 declines to say what everything else is. The measurement is the same under all three readings; the moral is not. If the group is independent, a CMDB is worth half what the naive comparison says. If it is tacit configuration knowledge, our +0.103 is a lower bound and the organisation already holds much of what the CMDB would tell it, unpaid for. The log cannot identify which is true, so we choose none of them. What survives either way is that the business case cannot be written against the intake block alone.

What we do not claim.. The reverse framing — that the item column “stands in for” the group — is not established. We measured it, by randomising items within groups, and obtained 44% of the item’s gain — but a routing-blind equal-size partition of items retains *more*, 56%, and one whose cell sizes follow the group’s own mass profile retains 25%. The leg is floor-dominated and we exclude it. Nor do we claim a direction of causation: the log records no counterfactual, so whether the item drives who logs the ticket or both follow a third cause is not identified here. Our claim is about what an analyst’s baseline choice does to a measured number, which does not depend on which is true.

Two fields we did not admit, and why.. Our baseline is not a terminus either. Hour of day and day of week are free and creation-time; adding both moves the item’s value from +0.103 to +0.099. `Service Component WBS (aff)` is harder: 100% populated, free, and admitting it takes the measured value to +0.023. We exclude it as near-deterministic in the item — only 58 of 2,929 items carry more than one value — so it is configuration data under another name. That criterion cuts both ways: 2,060 of 2,554 training items also map to a single opening group, though those items carry just 8.8% of incidents, so the high-volume ones are the multi-group ones. We know of no principled threshold that admits the group and excludes the service component. A reader who draws the line differently should read our headline as +0.023 —

instrument	$V(f \mid B_0)$	$V(f \mid B_1)$	reduction
ROC AUC	$+0.183$	$+0.103$	43.7% $[40.1, 47.1]$
Average precision	$+0.226$	$+0.090$	60.3% $[57.2, 63.4]$
Brier skill	$+0.168$	$+0.070$	58.4% $[54.4, 62.4]$
Nagelkerke R^2	$+0.223$	$+0.097$	56.6% $[52.8, 60.3]$
Net benefit, $\theta = 0.325$	$+0.093$	$+0.087$	6.3%
Net benefit, $\theta = 0.500$	$+0.084$	-0.016	119.2%

Table 3: The same reduction under four instruments that integrate over the operating range and two readings of one that does not. B_0 is the four intake fields; B_1 adds the opening group. Intervals are paired bootstrap, 2,000 draws, one resampled index set per draw shared by all four models. The net-benefit rows carry no interval on the ratio because the whole point of the row pair is that the ratio is not one number.

which is the decision this paper is about, and why we report a ladder rather than a number.

8. The Metric Axis

Section 6 varied B and held m at ROC AUC. This section holds B and the scores fixed and varies m. Same cohort, same split, same four fitted models, same seed: the only thing that changes is what is computed from the scores.

8.1. Seven instruments, six of them independent

Table 3 reports the reduction of Equation 2 under every instrument we can defend for this question. Detection at a fixed review capacity is kept in the table although Section 9.1 withdrew it as a headline, because a matrix of instruments is the right place to show what a tie-degenerate instrument does to a ratio.

Expected cost at a stated cost ratio is *not* a seventh axis. At cost ratio r the Bayes threshold is $1/(1+r)$, so the net-benefit weight $\theta/(1-\theta)$ is exactly $1/r$ and the cost avoided per case is $r \cdot \mathrm{NB}(1/(1+r))$ identically. The two therefore give the same reduction; we verified it over ten contrasts, where the two agree to machine precision, and report six instruments rather than seven.

The number we published is the smallest of them.. Across the four instruments that integrate over the operating range the reduction runs 43.7% to 60.3%, and ROC AUC sits at the bottom of that range. The previous version of this paper reported the AUC figure and treated the more principled instruments as a threat to it; on this data they are not, and a referee reading only Section 9.1 would have drawn the opposite conclusion. This is the second correction in this project's history that goes against the direction its errors usually take, and we record it as such rather than quietly banking it.

8.2. Why the instruments disagree

Three mechanisms are testable and we test all three. Each makes a prediction that can fail, and the first one did.

Aggregation, which is the largest.. AUC is $\int \mathrm{TPR}\, d\mathrm{FPR}$, so the increment a feature buys can be attributed to regions of the operating range without any modelling choice: the integral is additive over a partition of the FPR axis and the parts sum to the whole, which we assert rather than assume. We predicted the increment would be concentrated somewhere the net-benefit grid does not reach. It is not. Over the 31-point grid the group-aware model runs at every false-positive rate from 0.009 to 0.997, and the item's advantage is *diffuse*: the largest of ten equal FPR bands carries 18.2% of it and three bands are needed to reach half. AUC sums that whole profile; net benefit at one θ reads one point of it. The two are not disagreeing about a fact.

Baseline degeneracy, which explains where net benefit is lowest.. At $\theta = 0.200$, 0.300 and 0.325 the intake block acts on every incident in the test half: as a decision rule it *is* treat-all, and the group-aware baseline acts on 0.955 to 0.978 of them, barely distinguishable from it. An increment measured against a rule that does nothing comes out nearly the same on both rungs, and their ratio near one. Calling a baseline degenerate when it acts on more than 95% or fewer than 5% of arrivals — a threshold we declare before applying it — the reduction runs 0.047 to 0.435 across the 11 grid points where both baselines are degenerate and 0.581 to 1.168 across the 5 where neither is.

Calibration, which is real and provably not the whole story.. The item-aware model has calibration slope 1.040 and the intake block 1.391. Recalibrating on a held-out tail of the training half moves the net-benefit reduction by up to 0.253 where the denominator is safe and the proper-score reductions by up to 0.014. It moves the AUC and average precision reductions by exactly zero, because a monotone rescaling cannot change a rank statistic — a prediction that would have failed loudly had the recalibration not been monotone, and did not.

And a tie convention the two rank statistics do not share.. The intake block emits 23 distinct scores for 13,637 incidents. Average precision's default value equals its *negatives-first* bound, to machine precision, on all four models, because the precision-recall curve is evaluated where a tied block has been admitted whole; AUC's default equals a random tie-break to within 0.0004. Put both on the only convention a desk could implement, the reduction is 43.6% under AUC and 71.2% under average precision. The gap widens rather than closes, which is the outcome we did not expect and report because we did not.

9. What the Difference Buys: the Operating-Point Axis

An AUC difference is not a quantity a service desk can act on, so earlier versions of this paper stated the same comparison as detection at a fixed review capacity: rank the test incidents by predicted risk, take the top 5%, and count how many reassignment-bound incidents each model surfaces. That framing is withdrawn here, and the reason is the second of this round's two corrections.

9.1. Why the capacity framing was withdrawn

The paper already disclosed that the naive baseline cannot rank finely: four low-cardinality intake fields, one of them the redundant Priority column, emit only 23 distinct scores over 13,637 test incidents, so at 5% capacity just 47 rows sit strictly above the cut and the rest come from a tied block of 1,944. It said the factor was therefore partly about rank resolution rather than information, that the two had not been separated, and moved on. Separating them destroys the number.

The naive top-5% is a draw from a tie.. Of the 682 incidents the naive baseline nominates, 635 — 93.1% of the review budget — come from a single tied block, and that block is reassigned at 0.534 against 0.372 overall. How many of the right incidents the draw yields is therefore fixed by the block's own base rate and by nothing the model knows. The 47 rows the model does rank strictly above its cut contain 9 reassignment-bound incidents, a rate below

Tie-breaking policy	naive extra	honest extra	factor
random — the only implementable one	+271	+63	4.3
oracle: positives first in each tie	−26	+24	—
adversarial: negatives first	+608	+104	5.8

Table 4: Extra reassignment-bound incidents surfaced at a 5% review capacity by adding item identity, against a baseline that includes the opening group (honest) and one that omits it (naive), under three policies for ordering rows *inside* a tied block. The oracle and adversarial policies use the outcome and are unavailable to an analyst; together they bracket what tie-breaking alone can do. Medians of 400 paired bootstrap draws over test rows.

the base rate: at the top of its own ranking the naive baseline is worse than guessing. The group-aware baseline is in a different position, with 590 of 682 ranked strictly above the cut and only 92 drawn from a block of 246.

The factor's sign is set by the tie-break.. Table 4 prices that. Ordering rows inside the tied block optimally — which uses the outcome, so no analyst can — reverses the sign of the naive contrast: item identity surfaces 26 *fewer* incidents than the intake block alone, not 271 more. Ordering them adversarially gives +608. Nothing any model knows changes between those three rows. A quantity whose sign is set by how a coin lands inside a 1,944-row block is not a measurement of information, and we withdraw it.

A reader may object that the oracle policy favours the coarser arm by construction: it has the larger block to reorder, so it has the most to gain from an ordering it cannot obtain. That is the objection restated as the finding. A contrast whose magnitude is governed by the size of one arm's tie block, rather than by what either arm knows, is measuring the block. The random policy is the only implementable one and its value is the one an analyst would see; but its value is the expected yield of a draw from a block whose composition is a property of the cohort, not of the ranking, and the paper should not have read a difference in information off it.

The obvious repair is impossible.. It is natural to answer the objection by giving the naive baseline a finer representation, and we tried: cross-fitted target encoding of the intake block, per field and as a single composite key, on the same machinery used for the item column in Section 6. It cannot work. The four intake fields take 23 distinct combinations, so every row sharing a combination shares a score under *any* function of those fields, and combinations unseen in training collapse onto the training prior — the composite encoding emits 19 distinct scores, fewer than the one-hot model's, not more. The tie block is not an artifact of our estimator. It is what having four low-cardinality intake fields is.

What survives is narrower and we state it as such: the honest arm's extra catches are positive under every tie policy, including the oracle (+24 $[+9, +40]$), so knowing the affected item does surface more reassignment-bound incidents at this capacity than the group-aware baseline does. What does not survive is the ratio, and with it the claim that omitting the free field overstates the operational gain by more than it overstates the AUC gain.

9.2. Net benefit, which needs no capacity

Vickers and Elkin [26] supply the instrument the question actually needs. Fix a threshold probability p_t — the risk at which a desk judges a review worth doing — act on every incident scored at or above it, and compute

$$\mathrm{NB}(p_t) \;=\; \frac{\mathrm{TP}}{n} \;-\; \frac{\mathrm{FP}}{n} \cdot \frac{p_t}{1-p_t}.$$

The weight $p_t/(1-p_t)$ is the exchange rate the threshold implies: at $p_t = 0.25$ one missed reassignment-bound incident is worth three needless reviews. Net benefit is in true positives per incident, so we report $1000\times\mathrm{NB}$, which reads as reassignment-bound incidents caught per thousand arrivals net of the reviews spent catching them. It assumes no capacity, it is defined across the whole range of exchange rates a reader might hold rather than at one we picked, and — the point that matters after Section 9.1 — a threshold admits or excludes a whole tied block, so no tie is ever broken.

The scores are probabilities enough for this.. Net benefit is meaningless for a model whose scores are not calibrated, and a one-hot logistic model with 2,554 sparse columns at a fixed penalty is not obviously calibrated. Brier scores run 0.232 for the intake block, 0.206 with the opening group and 0.189 with the item; calibration slopes run 1.391, 1.185 and 1.040, so the intake block is the poorly calibrated arm and the item-aware model is close to ideal. Because a slope away from one moves net benefit, we repeat the whole analysis on scores recalibrated on a held-out tail of the training half — the model refitted on the first 85% of training, scored on the last 15%, and the scaling applied to test, with no test outcome entering it. No conclusion below changes.

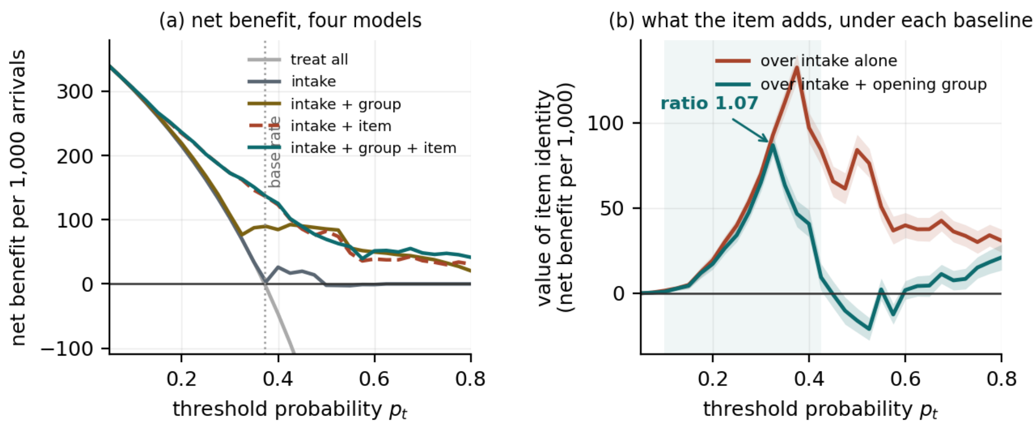

Figure 5: (a) Net benefit per 1,000 arrivals for the four models and for treating every arrival, across threshold probabilities. (b) What item identity adds under each baseline, with 2,000-draw paired bootstrap bands. The shaded region is where the group-aware increment's interval excludes zero. The two curves are indistinguishable until the threshold passes this cohort's base rate.

Where the item pays, and by how much less than it appears to.. Over a grid of 31 thresholds from 0.05 to 0.80, the group-aware increment is positive with its interval excluding zero at 20 points. Those 20 are not one run: 14 of them form a contiguous block from 0.100 to 0.425, the region either side of this cohort's base rate, and the remaining 6 sit at 0.675 and above, on the far side of the negative band described below. An earlier version of this sentence said all 20 lay in the run; every number in it was correct and the relation asserted between them was not (Section 15). The naive increment is positive and resolved at 29 of the 31. At the threshold where item identity is worth most, $p_t = 0.325$, it adds 86.9 reassignment-bound incidents per thousand arrivals over the group-aware baseline and 92.8 over the intake block: omitting the free field overstates its value there by a factor of 1.07. This is the number that replaces the withdrawn 4.3, and it is smaller than the ratio of 1.8 between the same two gains measured as AUC.

And where it does not pay at all.. Above the contiguous region the increment decays and then turns negative: at four grid points between 0.475 and 0.575 the group-aware increment's interval lies entirely below zero, and its minimum over the whole grid is -21.1 $[-27.7, -14.4]$ per thousand at $\theta = 0.525$. An earlier version gave the minimum as -16.1 $[-23.0, -8.9]$ at $\theta = 0.50$, which is the worst point among the five this paper had chosen to

p_t	over intake + group	over intake alone	ratio
0.20	+17.2 [14.0, 20.4]	+19.3 [16.0, 22.5]	1.12
0.30	+64.9 [59.8, 70.2]	+69.9 [64.7, 74.8]	1.08
0.40	+40.7 [33.0, 48.9]	+97.2 [88.6, 106.0]	2.39
0.50	−16.1 [−23.0, −8.9]	+84.1 [75.5, 93.2]	—
0.60	+1.8 [−3.7, 7.2]	+39.8 [32.5, 47.2]	—

Table 5: What item identity adds, as net benefit per 1,000 arrivals, under the two baselines. A ratio is not quoted where the group-aware increment's interval includes zero or lies below it, because a ratio whose denominator is crossing zero is not a quantity.

name rather than the worst point on the grid; the understatement was 31% and it flattered us (Section 15). At a one-for-one exchange rate, adding item identity to a model that already knows the opening group makes the desk worse off on this task. We have no mechanism to offer for that beyond the obvious one — a model with 2,554 item indicators can be confidently wrong where a coarse model abstains — and we report it because it is the operating region in which a desk that reviews a small share of arrivals is working, and because it is the honest reason the capacity framing produced the number it did. The earlier version measured the overstatement at the one operating point where its own group-aware arm had the least to give.

9.3. What the operational statement does not say

The threshold ladder of Section 6 is the operational statement of the same robustness, at the three target definitions reported there. None of this says a surfaced incident is a corrected one, and net benefit does not claim otherwise: it counts detections against reviews at a stated exchange rate, and what a detection is worth to a particular desk is a quantity this log cannot supply.

10. The Population Axis

Our item field is 100% populated. Real registers are not, and every practitioner who has read a draft of this paper has said so. That is a measurement, not a caveat, so we make it one: we mask the item column to a declared population level and re-measure the whole ladder, leaving both baselines untouched, so the curve is the value of a partially populated register rather than the value of a smaller dataset. A masked item becomes an explicit empty

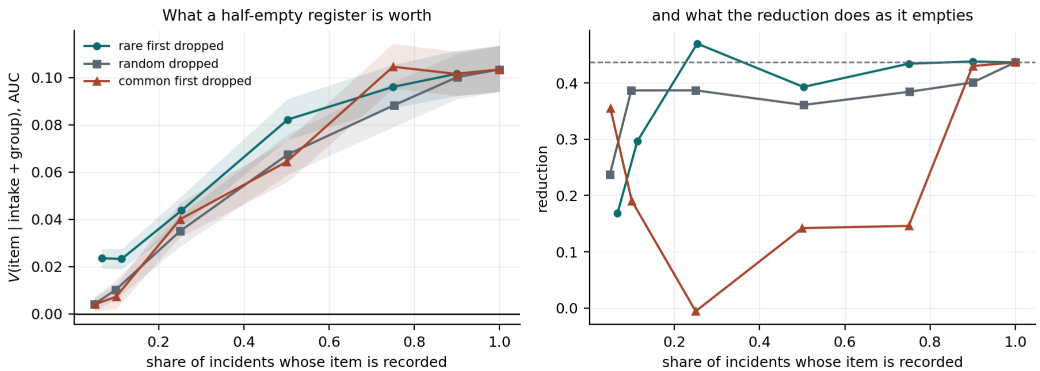

Figure 6: What a partially populated register is worth. Left: the item's value over intake plus the opening group, against the share of incidents whose item is recorded, under three degradation regimes. Right: the reduction of Equation 2 on the same axis. The estate is concentrated enough that 246 of 2,929 items carry 90% of the incidents, which is why dropping the tail costs so little.

level rather than a dropped row, because a desk with a half-empty register still has the incidents.

How an estate empties turns out to matter more than how empty it is, so we report three regimes: items masked independently at random; the least frequent items dropped whole, which is how a real estate degrades because discovery and manual curation cover the big central things first; and the most frequent dropped whole, which is not realistic and is reported as the other end of the range. At half population the group-aware increment is +0.082 when the long tail is missing, +0.067 at random, and +0.064 when the core is; the reduction at the same three points is 39.3%, 36.1% and 14.2%. Over the whole surface the reduction runs −0.6% to 47.1%, and its interval excludes zero at 19 of the 19 points measured. Figure 6 draws it.

The practical reading is that a buyer who prices a half-populated register from the random curve will be wrong in whichever direction their estate actually degrades, and by roughly a factor of three in the reduction. The usual caveat — "our figure is an upper bound because real data is worse" — is not supported on this axis either: at full population the measurement sits at the top of its own curve, so better discovery tooling cannot raise it.

11. The Corpus: Where the Effect Appears, and Where It Does Not

Everything above is one log. This section asks whether the admissibility effect is a property of process event logs in general or of ITSM data in particular, under a protocol registered before any of it was run.

11.1. Pre-registration

`PROTOCOL.md`, supplied as supplementary material and committed to the repository in a commit that adds no result file, fixes in advance: the two targets, both reported for every log; the rules that assign every attribute to exactly one of the high-cost entity f, the free per-event field g, the intake block B_0, a layer, or excluded; six exclusion codes; the estimator, split, seed and instruments; the log-level discriminators, computed before any model is fitted; and the outcome that would falsify the claim. Eight amendments were made after running the role assignment, which reads no outcome, and before fitting any model; each is recorded in that file with its reason, and every one is the registered text failing to implement its own stated intent. The check that they stop there is that after them the generic rules assign the primary log exactly the roles this paper assigns by hand — f the configuration item, g the opening group, B_0 the four intake fields, and the CI type and subtype as layers — although nothing in the rules names that log or those fields.

11.2. The corpus, and what the rules exclude

Twenty-three files across seven domains, fetched by DOI with a recorded SHA-256 for each; 22 logs parse. The registered rules admit 13 across six domains and exclude nine with a code. Two exclusions are worth stating as findings rather than as bookkeeping. In BPI Challenge 2012 and in all five BPI Challenge 2020 sub-logs, *every case is opened by the same actor*: the opening resource stamp has cardinality one, there is no free opening field to admit, and the question this paper asks cannot arise. In the Hospital Billing log the only reusable entity, `diagnosis`, is missing on 65% of first events and no other attribute reaches the registered cardinality floor.

11.3. The result, which is mostly negative

Nineteen log-target pairs survive the exclusion rules. Figure 7 shows every one where a reduction exists to be measured. On 10 of the 19 the entity is not resolvably worth anything over the intake block at all, so there is nothing for

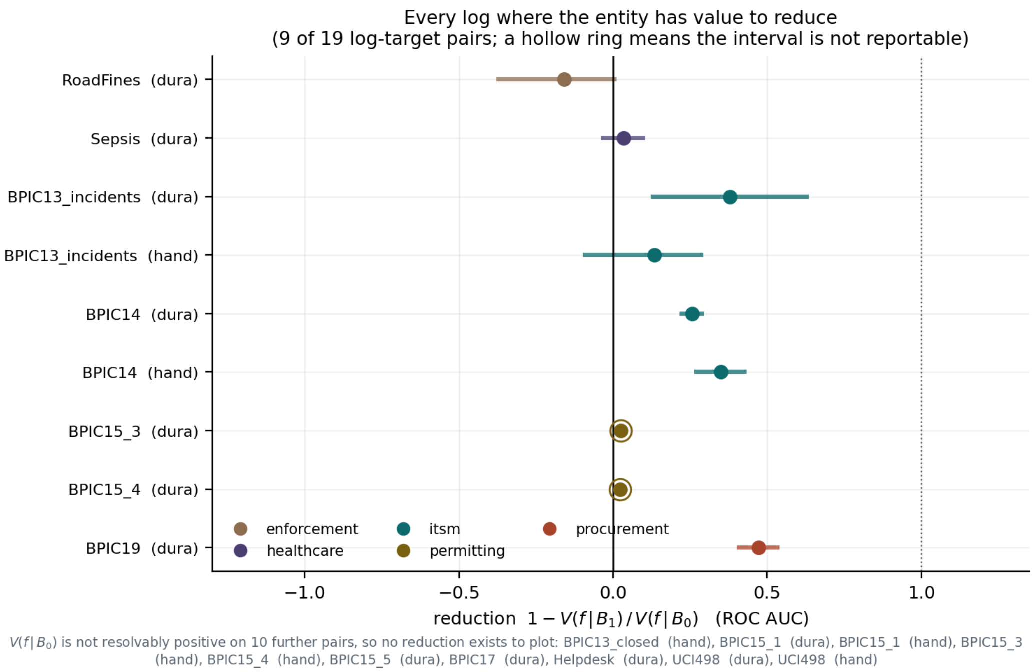

Figure 7: Every log-target pair where the entity is resolvably worth something over the intake block, so that a reduction exists. A hollow ring marks a pair whose reduction has no reportable interval because the denominator is not positive in every draw.

a free field to absorb and R is undefined rather than small. On the remaining nine the reduction's own interval excludes zero on four, drawn from three logs: the primary log on both targets, BPI Challenge 2013's incident log on the duration target (37.8% $[12.8, 62.8]$), and BPI Challenge 2019's procurement log on the duration target (47.1% $[40.8, 53.4]$ on 251,734 traces).

Eight of the 13 admitted logs are outside ITSM, and the reduction is resolvable on one of them. `PROTOCOL.md` §8 states that the claim is falsified if the effect appears on the two published ITSM logs and not on a majority of the admitted non-ITSM logs. It is falsified. We report that rather than reframing the claim to fit, and the paper's scope is bounded accordingly: what generalises is the *estimand* and the practice of reporting it as a surface, not the magnitude of any particular reduction.

11.4. Two things the corpus does establish

A precondition, not a moderator.. We attempted the prediction the plan for this round asked for: regress the reduction on log properties measurable before any model is fitted — entity cardinality, its concentration, the mutual information between f and g, trace length, whether g is a person or a team, and four more. On nine points with eleven predictors the leave-one-out R^2 is -1.323 against a permutation null of 2,000 shuffles at $p = 0.448$: the fit predicts a held-out reduction worse than its own mean does. We report that and show no in-sample fit in its place. What the corpus does show is coarser and more useful: the effect requires an entity that predicts the outcome at all, and on most public logs it does not.

A generic target is not the published target, and we checked.. The protocol's handover target counts distinct opening groups in a trace. The primary log ships its own reassignment count, which is what this paper's other sections use. They are not the same construction and they do not agree: the generic target fires on 92.7% of incidents against the published target's 41.1%, and the two agree on 46.0% of them, barely above what independence between two marginals of that size would give. The tautology control of `PROTOCOL.md` §4.3 splits with them — under the generic target the largest opening group has the higher rate, under the published target the lower. The corpus measures a generic workflow outcome; it does not measure this paper's task on twelve further organisations, and it would have been easy to write as though it did.

12. A Reporting Standard

We propose that a feature-value claim be reported as a surface over the arguments of Equation 1 rather than as a point, and that the minimum reportable form is:

1. **The baseline ladder.** Every already-recorded field admitted to B, named, with the admissibility criterion stated before it is applied, and V at each rung. A field the organisation already holds and does not admit is a decision, and its cost is the difference between two rungs.
2. **At least one integrating and one operating-point instrument.** On this data the two families differ by more than the effect. Reporting only one is reporting one of two answers without saying which question was asked.
3. **The operating range, not a chosen point.** If the claim is decision-analytic, the whole curve with intervals, and explicitly the region where the increment is not resolvable or is negative.

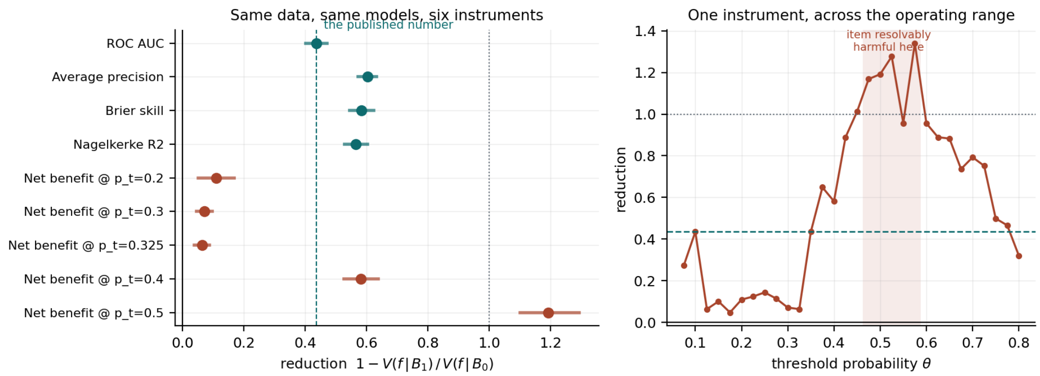

Figure 8: Left: the reduction under six instruments, paired bootstrap intervals, the published AUC figure dashed. Right: the same reduction under one instrument across the operating range. The shaded band is where the item's increment over a group-aware baseline is resolvably negative.

4. **The register's population, if the feature is a lookup.** With the degradation regime named, because the regime moves the answer by more than the level does.

Figure 9 is that report for one instrument on this data, and Figure 8 is the same claim across instruments. Neither figure contains a number that is not in a result file, and every number in both is re-derived from those files by the checker described in Section 15.

13. The Question We Could Not Answer, Answered

The previous version of this paper had a section here titled "One Thing We Could Not Establish". A third field, the knowledge-article reference, sits on the same `Open` row, is 100% populated, and costs nothing. Adding it to the baseline raises AUC to 0.805 and drives the measured value of item identity to -0.003. Whether that figure belonged in the headline depended on whether the field is available at creation, and we could not determine it from the three files we held. That section named the missing evidence: an interaction detail file shipped in the same public collection, which we had not obtained.

We obtained it. `Detail_Interaction.csv` records 147,004 service-desk interactions.

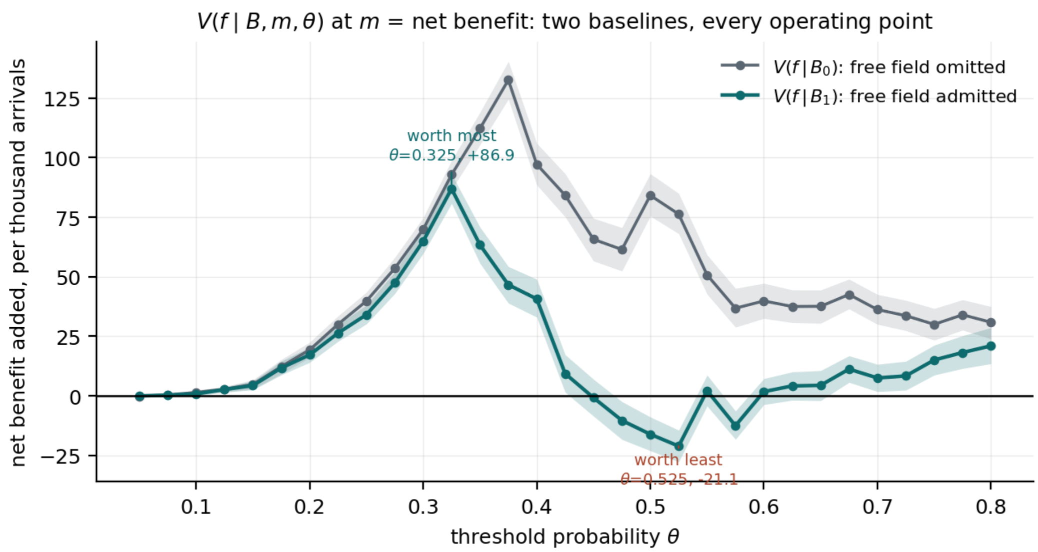

Figure 9: $V(f \mid B, m, \theta)$ at m = net benefit, both baselines, every operating point, with paired bootstrap bands. A single number for "what the item is worth" is a choice of one x-coordinate and one of these two curves.

What it settles.. 94,250 of those interactions — 64.1% of the file — never produce an incident at all, and 99.7% of those were resolved on the first call. Every one of them carries a knowledge reference, across 1,978 distinct articles. A field the incident process assigns cannot be populated on records the incident process never touches, so the reference is written by the service desk during the call. Of the incidents that do join an interaction, 92.4% of the cohort, the interaction opened before the incident in 100.00% of cases with a median gap of six minutes, and the desk's recorded handling time on it had elapsed before the incident opened in 98.6%.

What it does not settle, and we say so.. Agreement with the closed record still cannot discriminate. On the 41,413 incidents whose interaction was worked to completion first, the knowledge reference agrees with its interaction's value on 100.00% and the closure code — which is certainly not creation-time — on 98.97%. The reasoning of the previous version stands: an interaction is itself a closed record. The subset whose interaction had formally *closed* before the incident opened contains five incidents, and nothing is concluded from five.

The consequence, which removes our headline.. Admitting the reference carried by an interaction that was worked to completion before the incident

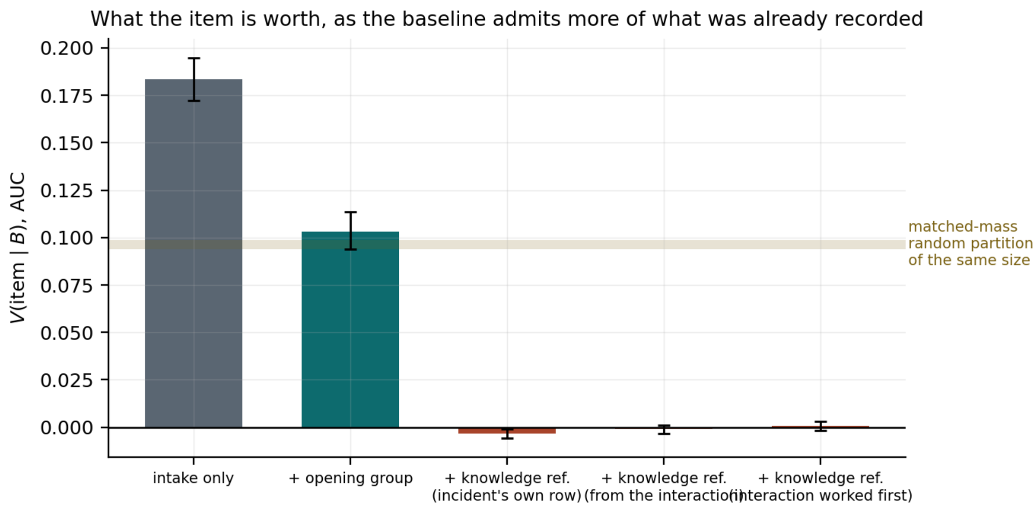

Figure 10: What item identity is worth as the baseline admits more of what the organisation already recorded. The shaded band is the range left by five matched-mass random partitions of the same cardinality as the knowledge reference.

opened — populated on 91.1% of the cohort — takes the item's measured value from +0.103 to +0.001 [−0.002, +0.003], a reduction of 99.7% against the intake-only baseline rather than 43.7%. Figure 10 draws the ladder.

Two nulls, because a result of that size is not believed on sight.. A collapse to zero could be collinearity — the reference being item identity relabelled — or dimensionality, 1,700 extra sparse columns at fixed penalty being a burden the item must then overcome. Neither survives. 78.8% of knowledge articles map to exactly one item, against 80.7% for the opening group, which absorbs less than half of the item's value rather than all of it; so determinism at that level does not collapse a marginal. And five matched-mass random partitions of the same cardinality — built by slicing a permutation rather than by `searchsorted` on cumulative mass, which is the bug that killed correction three — reach base AUC at most 0.6479 and leave the item worth +0.094 to +0.099. The real field reaches 0.8041 and leaves it worth +0.001.

What this does to the paper.. Our thesis is that the measured value of a field is set by an admission decision the analyst makes silently. The strongest available demonstration of that thesis is that it removes our own headline, and this is that demonstration. We do not claim the knowledge reference is *proved* creation-time; we claim the balance of evidence has moved decisively, that a paper whose headline is contingent on excluding a field must report what admitting it does, and that +0.103 should be read as the value of item identity *given that one particular free field is admitted and another is not.* Section 12 is the general form of that sentence.

14. Limitations

Era, decomposed into four measurements.. The primary log is six months of 2013 and 2014 from one estate, and practice has moved. "Practice has moved" is not one claim, however, and each of its four usual components has a proxy in data we hold. *(i) Discovery auto-populates the estate.* Ours is already 100% populated, so the measurement sits at the top of its own curve and better tooling cannot raise it; what a modern estate changes is which part of the register is populated, and Section 10 measures that. *(ii) Incidents arrive pre-stamped.* In 2014, 97.7% of incidents that came through the service desk already carried the item their originating call carried. A change that pre-stamps arrivals is a change from 97.7% to at most 100%. *(iii) Service mapping rather than item identity.* The service component — 272 values in this cohort — reaches +0.078 over intake plus group against the item's +0.103, so it captures 75.3% of it, and the item's marginal over it is +0.023. *(iv) More intake channels, fewer central desks.* This is the largest sensitivity in the paper. Subsampling the cohort so that the largest opening group holds between 0% and 95% of it moves the reduction from 95.3% to 6.3%: when one desk opens everything the opening group carries almost no information and absorbs almost nothing, and when opening is distributed it absorbs nearly all of the item's apparent value. A 2026 organisation with more channels sits at the end of that sweep where the free field matters *more*, not less.

Data quality.. Section 10 replaces the caveat this paragraph used to make. What remains is that we measure population and not *accuracy*: a register that is fully populated with wrong values is not modelled here, and we have no way to detect one in a public export.

Free text, and a search rather than a regret.. Neither of the published logs carries a usable short description, and every real deployment has one. Kapel et al. [11] rank two free-text fields above CI Name for a related target. We therefore searched: every attribute of every one of the 22 parsed logs, against a criterion declared in `PROTOCOL.md` §7 before the search ran. Of 596 attributes, zero satisfy it. The field a referee will name, UCI 498's `u_symptom`, is 3.4% unique with a mean length of 10.9 characters: a coded symptom, not a description. Two amendments were needed to stop the criterion admitting formatted identifiers, and they are recorded. The threat is real, it is the most important open question this study leaves, and it cannot be tested on public process-mining data. We report the search rather than the regret.

Construct and mechanism.. Reassignment is a proxy for misrouting and also fires on legitimate escalation; the within-item shuffle preserves item membership but not other correlates of identity, so it bounds the proxy share rather than isolating it; and the opening group's distribution drifts between our halves. After round sixteen's withdrawals, we do not claim a measured direction for the mechanism.

The corpus measures a different target from the rest of the paper.. Section 11 applies a generic handover target so that one rule can run on twenty-two logs. On the primary log that target and the log's own reassignment count agree on 46.0% of incidents. The corpus therefore bounds the generality of the *estimand and the practice*, not of this paper's particular magnitude, and we say so there rather than letting thirteen log names imply thirteen replications.

What is still outside reach.. No deployment, no organisational partner and no practitioner validation; a single author with no institutional affiliation; public benchmark data whose newest log is from 2019. None of the four is fixable by further analysis and we would rather state them than have them inferred.

The operational statement is a detection statement.. Net benefit counts detections against reviews at a stated exchange rate. It does not say that a detected incident is a corrected one, that a desk given the warning would route differently, or what a correct routing is worth. Section 9 also reports a region of thresholds in which the item is worth nothing or less than nothing over the group-aware baseline, and we have no mechanism for that beyond over-confident predictions from a high-dimensional model.

15. Corrections

Eleven errors of our own are reported as results rather than edited away, because each is an instance of the failure this paper is about — a control, a claim or a baseline asserted rather than checked — and because the pattern across them is more informative than any one.

1. **A null that could not fail.** The mechanism's floor was originally drawn by assigning cell labels per *row*, so two incidents sharing an item landed in different cells and the association was destroyed by construction. The null could only return zero, and we published a margin of 89 points against it (Section 7).
2. **The same null, rebuilt, was still a knob.** Rebuilt at item level it was swept only to 800 cells while the leg it bounds uses 2,929, and the reported margin was taken at the cell count that maximised it. At matched granularity the margin is 3.5 points at $z = 0.9$ and we now withdraw it. This is the fifth time a null in this project has been drawn at the wrong level.
3. **An asymmetry that was an algebraic identity.** The estimator-free evidence for the direction of the mechanism was a ratio of two uncertainty coefficients, which is identically the ratio of two marginal entropies. It was published for eight rounds with both entropies sitting in the same results file, one division away.
4. **A factor printed without an interval.** The capacity-table factor was first given as a point estimate, on an unmatched pair of models, in a paper that elsewhere refuses to resolve +0.0017 past two decimals.
5. **A central field whose meaning we asserted.** The opening group was described as the routing queue for several versions. It is the group that logged the incident (Section 4).
6. **A claim about a public dataset that our own repository refuted.** We wrote that the third public ITSM log has no affected-item field. It has one, populated on every case, and a script in our own repository had been treating it as such (Section 4.2). This was the sole support for describing the study as single-organisation of necessity, which we had presented as a constraint rather than a choice. It was a choice.
7. **An operational factor whose sign was a tie-break.** Correction four gave that factor an interval; this round asked what it measured. At a 5% review capacity 93.1% of what the naive baseline nominates is drawn from one tied block, and reordering rows inside that block — which changes nothing any model knows — moves the naive arm from −26 to +608. We withdraw the factor and rebuild Section 9 on net

benefit, which never breaks a tie. The replacement number is 1.07, not 4.3 (Section 9.1).

8. **A control run on one of the two rungs it bounds.** The shuffled-item encoder null was run on the group-aware rung only, while the reduction it is meant to bound is computed from two rungs, and its residual under boosting is eleven standard errors from zero rather than noise. Run on both rungs, the correction moves the boosting reduction by 3.4 percentage points, and upward (Section 6).
9. **An extremum that was the worst of the points we had named.** We wrote that the group-aware increment turns negative, "reaching -16.1 $[-23.0, -8.9]$ per thousand at $p_t = 0.50$". The word *reaching* names an extremum. The extremum is -21.1 $[-27.7, -14.4]$ at $\theta = 0.525$, which is 31% larger; 0.50 was in the list of five thresholds our own table happened to name and 0.525 was not. Every literal in that sentence was correct and was computed from a file already in the repository (Section 9.2).
10. **A count and a run reported as the same set.** We wrote that the increment is resolvably positive "at 20 points, in a contiguous run from 0.100 to 0.425". Fourteen are in that run; six sit at 0.675 and above. Counting a set and describing an interval are different operations and this sentence performed one and reported the other (Section 9.2).
11. **An open question we left open with the evidence one download away.** The previous version reported that it could not establish whether the knowledge-article reference is available at incident creation, named the file that would settle where the field comes from, and did not obtain it. The file is in the same public collection as the three we used. Obtaining it shows the reference is written by the service desk on calls that never become incidents at all, and admitting it takes our headline from +0.103 to +0.001 (Section 13). We record this as a correction rather than as progress because the paper's own thesis is that an unexamined admission decision sets the answer, and we left ours unexamined for a round.

Eight of the eleven flattered the result, one — the dataset claim — excused its principal limitation, one would have made the result larger had we noticed it earlier, and one removed the result altogether. We draw the obvious inference about the direction in which unchecked assertions travel.

Two of the three found this round are worth separating out, because they are a kind the apparatus was built to catch and could not. The paper is checked by a program that re-derives every numeric literal from a result file, and a corruption suite of 149 mutations that has found eight holes in that program. Every literal in corrections nine and ten passed: -16.1, -23.0, -8.9, 0.50, 20, 0.100 and 0.425 are all correct values sitting in correct places. What was wrong was the relation the prose asserted between them — an extremum in the first case, a set-equals-interval identification in the second. A checker that verifies numbers and not the words joining them will certify both. The checker now compares the named extremum against the extremum and the stated count against the run length, and the general lesson — that a directional word is a claim and needs a check of its own — is the one this project keeps relearning.

16. Conclusion

What a recorded field is worth is not a number. It is $V(f \mid B, m, \theta)$, and on this decision every one of those arguments moves the answer by more than the effect we set out to report.

Vary the baseline: knowing which configuration item an incident concerns is worth +0.183 AUC against four intake fields, +0.103 once one further field the organisation already records is admitted, and +0.001 $[-0.002, +0.003]$ once a second is. Vary the metric on those same scores: the reduction runs 43.7% to 60.3% over six instruments, and the figure we previously published is the smallest of them. Vary the operating point: it runs 6.3% to 119.2%, and in a band above the base rate the item is resolvably harmful. Vary the register's population and the regime by which it empties: at half population the item is worth +0.082 or +0.064 depending on whether the tail or the core is missing. Vary how the organisation takes work in: across intake mixes the reduction runs 6.3% to 95.3%.

None of those variations is a defect in any instrument or an error in any measurement. Each is correct about the question it asks. The point is that a single number answers one of those questions without saying which, and that the person writing the business case is the person choosing.

We tried to make the finding general and it did not go. A pre-registered protocol over 22 public logs admits 13 and finds a resolvable reduction on three; the registered generality claim is falsified and Section 11 reports it. What the corpus does establish is the precondition: the effect needs an entity that predicts the outcome at all, and on most public process logs there is not one. What we offer in its place is Section 12: a minimum reportable form for an incremental-value claim, demonstrated on data where following it removes the paper's own headline. That is the strongest evidence we can give that it is worth following.

Acknowledgements and Reproducibility

The manuscript was subjected to seventeen rounds of adversarial review, the later rounds machine-assisted, in which each round was asked to break the previous version's claims; the corrections in Section 15 came from that process. The protocol of Section 11 was registered in the repository before the analyses it governs were run, in a commit that adds no result file. The author set every research question, adjudicated every proposed correction against the data, and is responsible for the content. The datasets are public [8, 22]; code, figures and a checker that recomputes every quantity here: https://github.com/sudhirdixit1/cmdb-field-admission.

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